

DFR-FORENSICS v2.8.0

Disk & File Resurrection: Low-Level Architecture & Technical Whitepaper

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1. Forensic Problem Statement & Architectural Philosophy

In modern cyberattacks (destructive wipers like *HermeticWiper*, *CaddyWiper*, *WhisperGate*) and catastrophic storage hardware incidents (initial bad sector bursts, power loss during partitioning), storage boot structures are systematically targeted. When sector LBA 0 (MBR) or LBA 1 (GPT) is wiped with zeroes, operating systems (Windows, Linux, macOS) and traditional commercial forensic suites flag the media as **completely unallocated or uninitialized**.

DFR-Forensics (*Disk & File Resurrection*) was engineered by **Dam-FOR3K** as an autonomous, professional-grade, pure-Python forensic platform operating without mandatory external closed-source DLLs. It enforces strict digital evidence principles:

- **Strict Zero-Write Policy:** No write operations are ever performed on the target evidence. Raw disk images (RAW, DD, VMDK, VHD/VHDX, QCOW2, E01, AD1, AFF4, DMG) and live physical drives (*ll.PhysicalDriveX*) are mounted exclusively with hardware-level read-only flags.
- **Virtual Copy-On-Write (COW) Architecture:** All geometry repairs, volume table syntheses, cryptographic container decryptions (BitLocker, LUKS), and superblock restorations execute purely in an in-memory virtual abstraction layer, ensuring 100% evidentiary integrity.
- **Mathematical Reconstruction via Invariants:** No offsets or cluster sizes are hardcoded. The engine derives architecture from universal filesystem invariants (media descriptors, cluster chain relationships, and magic headers) to recalculate exact layout even when primary headers are wiped.
- **Universal Interactive & Headless Workflow:** Combining an interactive PySide6 graphical interface with a fully scriptable CLI for CSIRT/SOC incident response pipelines, the suite unifies spatial layout mapping, Binwalk-inspired entropy tracking, Slack Space inspection, multi-threaded raw pattern search, and tree-based virtual file extraction across all major desktop and embedded OS families.

2. Low-Level Partition Table Architecture

A. Master Boot Record (MBR) & EBR Chains

Sector LBA 0 hosts the legacy MBR: 446 bytes of bootstrap code, 4 partition descriptor entries of 16 bytes each (offsets 446 to 509), and the boot signature 0x55AA (offsets 510-511). Each entry stores the boot flag, filesystem type ID (e.g. 0x07 NTFS/exFAT, 0x83 Linux Native), starting LBA, and total sector count. When extended partitions are used, the tool recursively navigates the linked chain of **Extended Boot Records (EBR)**.

B. UEFI GPT Specification (GUID Partition Table 2.10)

Modern storage adheres to the UEFI GPT layout:

- **LBA 0 (Protective MBR):** Dummy MBR containing a single partition of type 0xEE spanning the entire drive to prevent legacy utilities from misidentifying the disk.
- **LBA 1 (Primary GPT Header):** 92-byte header beginning with signature EFI PART (0x5452415020494645). Stores disk GUIDs, partition table pointer (LBA 2), backup header pointer (LBA N-1), and two independent CRC32 checksums.
- **LBA 2 to 33 (Partition Entry Array):** 128 entries of 128 bytes each (16,384 bytes). Each record specifies partition type GUID, unique GUID, first LBA, last LBA, attributes, and UTF-16LE partition label.
- **LBA N-33 to N-2 (Backup Array) and LBA N-1 (Backup Header):** Full replica at the end of the disk providing fault tolerance.

C. CRC32 Recalculation Algorithm (IEEE 802.3)

When a wiper zeroes out the first megabytes of a disk (LBA 0..2048), the Primary GPT is wiped, but the Secondary GPT (LBA N-1) is frequently intact. DFR-Forensics restores the structure as follows:

1. Extract partition array from LBA N-33 and map to LBA 2..33.
2. Invert geometry pointers: $\text{CurrentLBA} = 1$, $\text{BackupLBA} = N-1$, $\text{PartitionEntriesLBA} = 2$.
3. Recalculate CRC32 of partition array (16,384 bytes) using reflected polynomial 0xEDB88320.

4. Reset the 4 bytes of Header CRC32 (offset 16) to zero, calculate CRC32 over the 92-byte header, and inject back.
5. Synthesize a valid Protective MBR at LBA 0.

3. Standard Filesystem Engines & Autonomous Recovery

A. FAT12 / FAT16 / FAT32 & Backup Boot Sector Failover

In FAT filesystems, the **BPB (BIOS Parameter Block)** in the Volume Boot Record (LBA 0) dictates all cluster geometry: bytes per sector, reserved sectors, number of FAT tables, root entries, and sectors per cluster (SPC). When LBA 0 is zeroed out by a wiper, **DFR-Forensics** applies a dual-tier resilience strategy:

- **Backup Boot Sector Failover (LBA 6):** Under FAT32, the engine probes sector 6 (official OEM boot sector backup). If intact, original volume geometry is restored instantly without heuristic derivation.
- **Mathematical Orphan BPB Reconstruction:** When both sectors 0 and 6 are destroyed:
 1. Scans media descriptor bytes (0xF8 / 0xF0) to locate FAT1.
 2. Computes: **Sectors Per FAT = LBA(FAT2) - LBA(FAT1)**.
 3. Positions the Root Directory table immediately following FAT2.
 4. Dynamically solves cluster size (SPC) by testing candidate offsets against real file signatures (JPEG, PDF, ZIP).

B. Linux EXT2 / EXT3 / EXT4 & Extents Tree Architecture

Linux EXT organizes storage into Block Groups. When the primary superblock is wiped (LBA 2) or files are fragmented:

- **Sparse Superblock Discovery:** Automatically sweeps offset 8,389,632 (LBA 16,386) and group boundaries up to 32 MB to detect magic 0xEF53 and restore group descriptors.
- **Full EXT4 Extents Support (magic 0xF30A):** Native traversal of the extent tree stored within modern 60-byte inode blocks. Recursively decodes index nodes (depth > 0) and leaf extent descriptors (ee_block, ee_start, ee_len), ensuring bit-exact reassembly of multi-gigabyte fragmented files.
- **Direct, Indirect & Double Indirect Pointer Resolution:** Chirurgical traversal of legacy Ext2/3 block trees, skipping 1,024-byte pointer metadata blocks interspersed throughout large files.
- **Directory Slack Space Undelete:** Unlinked file records absorbed by active entries' rec_len are extracted with original names and metadata.

C. Windows NTFS (\$MFT, \$MFTMirr & Backup VBR)

Direct pure-Python **\$MFT** parsing with critical failover mechanisms:

- **\$MFTMirr Failover (cluster offset 0x38):** When Record 0 of the primary \$MFT is damaged or wiped, the engine fails over to \$MFTMirr to recover non-resident \$DATA runlists mapping the entire MFT across disk clusters.
- **Backup VBR Resolution (Last Sector LBA N-1):** When the partition boot sector is blank, BPB parameters (cluster size, record size, MFT cluster) are retrieved from the volume end.
- **Transaction Log Carving (\$LogFile) & Undelete:** Comprehensive reconstruction of active and deleted files with quadruple MACB timestamps (\$STANDARD_INFORMATION).

D. Native exFAT Engine (Main & Backup VBR)

Pure-Python forensic parser for flash and external storage media: Backup VBR failover (sector 12), directory entry chain parsing (primary active 0x85 / deleted 0x05, stream extension 0xC0 with contiguous allocation flag NoFatChain, and Unicode name sequences 0xC1). Direct bit-exact extraction without third-party drivers.

4. Embedded, Mobile & Flash Filesystems (v2.8.0 Additions)

Version 2.8.0 expands DFR-Forensics beyond traditional PC storage into industrial firmware dumps, automotive infotainment systems (IVI/ECUs), mobile devices, and modern Apple architectures:

A. QNX Flash Filesystem (F3S / ETFS) - Automotive & Industrial IoT

In automotive telematics and critical infrastructure (BlackBerry QNX Neutrino), raw NOR and NAND flash memories frequently use **QNX F3S (Flash Filesystem v3)** or **ETFS (Embedded Transaction FS)** instead of partition tables. Physical flash is organized in *Erase Units* (64 KB, 128 KB, or 256 KB blocks):

- **Erase Unit Headers (magic 0x66)**: Identification and parsing of block control records managing block wear leveling and sequence numbers.
- **File & Extent Headers (magic 0x76)**: Sequential parsing of metadata records storing UTF-8 filenames, file sizes, POSIX permissions, UID/GID, and payload extents.
- **In-Place Update Reconstruction**: Because F3S writes updates across new flash units rather than overwriting in place, the parser resolves current file versions while also uncovering historical, superseded versions for deleted data recovery.

B. Apple HFS+ / HFSX & \$AllocationFile Bitmap Carving

Although APFS is primary on recent macOS releases, HFS+ (Hierarchical File System Plus) remains widespread on external drives, Time Machine backups, and legacy forensic images:

- **Volume Header at Offset 1024 (magics 'H+' 0x482B and 'HX' 0x4858)**: Extraction of allocation block sizes, total block counts, and extent descriptors for system metadata files.
- **Catalog B-Tree Traversal**: Pure-Python balanced B-Tree parser (header nodes, index nodes, leaf nodes) mapping Catalog Node IDs (CNID) to data forks, resource forks, and HFS UTC timestamps.
- **Targeted Carving via \$AllocationFile Bitmap**: Decodes the allocation bitmap file (1 bit per allocation block). DFR-Forensics maps exact unallocated block intervals, accelerating carving scans and eliminating healthy file false positives.

C. Apple APFS (Apple File System) & Separated Volume Discovery

In APFS architectures, a single physical container (GUID 7C3457EF-0000-11AA-AA11-00306543ECAC) hosts multiple logically independent volumes (System, Data, Preboot, Recovery, VM) sharing free space dynamically:

- **Object Map (OMAP) & B-Tree Navigation**: Resolves virtual object IDs (OID) to physical block addresses (PBA).
- **Individual Sub-Volume Discovery**: As of v2.8.0, APFS sub-volumes are automatically discovered and displayed separately in the partition table (as indented child entries) and in the virtual file explorer dropdown, enabling analysts to investigate user data independently from sealed system volumes.

D. Embedded Linux Filesystems: SquashFS, CPIO, F2FS, EROFS, UBI/UBIFS

DFR-Forensics v2.8.0 integrates specialized parsers for embedded Linux distributions and mobile dumps:

- **SquashFS v4 (magic 'hsqs' 0x73717368)**: High-density compressed read-only filesystem found in routers, gateways, and IoT firmware. Decompresses metadata tables, directory entries, and data fragments (Zlib, LZ4, Zstandard, XZ).
- **CPIO / Initramfs (SVR4 portable formats '070701' and '070702' with CRC)**: Decodes Linux boot ramdisk archives, extracting init scripts, kernel modules, and embedded payloads.
- **F2FS (Flash-Friendly File System, magic 0xF2F52010)**: Superblock probing at offsets 1024 and 5120 bytes, checkpoint status validation, and segment layout discovery for modern Android devices.
- **EROFS (Enhanced Read-Only FS, magic 0xE0F5E1E2)**: Probes superblock at offset 1024, traversing uncompressed inodes and metadata on modern Android/Huawei vendor partitions.
- **UBI / UBIFS (Unsorted Block Images)**: Decodes Erase Counter headers ('UBI#' 0x55424923) and Volume ID headers ('UBI!' 0x55424921) on raw flash media without hardware FTL.

5. Volume Shadow Copies (VSS), File Slack Inspection & Raw Search

A. Windows Volume Shadow Copies (VSS Snapshots)

On NTFS partitions, the Windows VSS subsystem maintains point-in-time snapshots of filesystem states, preserving deleted files, previous registry hives (SAM, SYSTEM, SOFTWARE), and event logs:

- **Heuristic VSS Catalog Sweeper (magic 'scek' 0x7363656B):** Sweeps metadata storage chunks within NTFS volume structures.
- **Snapshot Descriptor Decoding:** Extracts snapshot GUIDs, Windows FILETIME timestamps (100 ns intervals since Jan 1, 1601), and volume target associations.
- **Direct Explorer Integration:** Discovered snapshots appear directly in the virtual file explorer dropdown, allowing incident responders to inspect historical filesystem states before malware execution or anti-forensic wiping.

B. Forensic File Slack Inspector (RAM Slack & Drive Slack)

When a file with logical size L is written to disk, the operating system allocates an integer number of clusters (physical allocated size $P = \lceil L / C \rceil * C$). The residual bytes between L and P constitute **File Slack**, a critical goldmine for digital investigators:

- **RAM Slack (Last Sector Remainder):** The gap between the logical end of file and the end of the current 512-byte physical sector. On legacy operating systems, this region was padded with transient RAM buffer artifacts (passwords, email fragments, encryption keys). Modern OSs zero this area.
- **Drive Slack / Volume Slack (Remaining Cluster Sectors):** The remaining unused sectors within the allocated cluster. This space contains uninitialized data from previous deleted files that previously occupied those sectors (deleted documents, malware fragments, residual plaintext).
- **Right-Click Context Inspection & Extraction:** In the virtual file explorer, right-clicking any file opens the Slack Space Inspector, displaying hexadecimal previews and providing one-click export for secondary carving or steganographic review.

C. Multi-Threaded Streaming Raw Keyword & Regex Search

To search for indicators of compromise (IOCs), case-specific keywords, or regular expressions without relying on filesystem catalog integrity, DFR-Forensics provides a multi-threaded streaming chunk scanner:

- **Multi-Encoding Support:** Concurrently sweeps ASCII, UTF-8, and UTF-16LE (Windows Unicode).
- **Regular Expressions & Overlap Handling:** Full regex matching across chunks with sliding window overlap to guarantee signatures spanning chunk boundaries are never missed.
- **Detailed Reporting & CSV Export:** Every match is indexed by exact physical byte offset and LBA sector, with surrounding context snippets and instant CSV export.

D. Binwalk-Inspired Flash Entropy & 0xFF Unprogrammed Handling

On raw flash chips (NAND/NOR, Chip-Off dumps), unwritten or erased blocks are filled with **0xFF** bytes (unprogrammed cell state) rather than zeroes (0x00). DFR-Forensics adjusts its visual entropy mapper to discriminate 0xFF flash space (grey lines), clear code/text (green), compressed media (blue), and encrypted volumes (deep purple).

6. Intelligent Carving Engine, NIST CFTT & BraidResolver

Unlike naive carvers that extract arbitrary byte spans resulting in false positives, **DFR-Forensics** applies strict format-specific structural and mathematical validation:

Format	Header Signature	Validation Algorithm & Length Computation
JPEG / JFIF	FF D8 FF E0..FE	Sequential marker parsing (SOF, DQT, DHT, SOS). MCU block validation and strict search for EOI marker FF D9.
PNG	89 50 4E 47 0D 0A 1A 0A	Full CRC-32 (IEEE 802.3) integrity verification on each chunk (IHDR, IDAT, IEND). File boundary locked at the final IEND chunk byte.
BMP	42 4D ('BM')	Little-endian total filesize verification from bytes 2..5. Non-negative dimensions and valid DIB header validation.
GIF	GIF87a / GIF89a	Logical Screen Descriptor dimensions check, full block structure traversal (extensions 0x21, image descriptors 0x2C, LZW sub-blocks) down to the legitimate trailer 0x3B for byte-exact boundary determination.
OLE CFBF (DOC, XLS, PPT)	D0 CF 11 E0 A1 B1 1A E1	Internal OLE FAT traversal to compute exact sector allocation and avoid truncated documents.
PDF	%PDF-1.x	Bounded forward scan for the nearest %%EOF marker to prevent document concatenation.
TIFF 6.0	49 49 2A 00 ('II*\x00') 4D 4D 00 2A ('MM\x00*')	Full Little-Endian and Big-Endian support. Traverses Image File Directory (IFD), computes exact physical span via Strip/Tile ByteCounts, and filters embedded JPEG EXIF false positives.
ZIP / Office XML	50 4B 03 04 ('PK..')	Local File Header chaining and validation via Central Directory (PK\x01\x02) and EOCD (PK\x05\x06).

Strategic Sector Alignment & NIST CFTT Compliance

- **Flexible Sector Alignment (512B, 4096B, 1B unaligned):** High-speed default mode aligned on 512-byte physical sectors, or 1-byte exhaustive scanning for unaligned memory dumps.
- **Unallocated Space Only:** Probes filesystem bitmaps directly (NTFS \$Bitmap, FAT tables, exFAT bitmap, EXT bitmaps, APFS Spaceman, HFS+ \$AllocationFile), accelerating carving 5x-10x.
- **BraidResolver De-Braiding (NIST CFTT Graphic-Braid):** Successfully disentangles interleaved file pairs [1A, 1B, 2A, 2B] with verified in-memory pixel rendering, achieving 100% pass on the NIST suite.
- **Visual Resilience Cocktail:** Auto-closure of truncated streams (virtual FF D9), permissive decoding (Pillow LOAD_TRUNCATED_IMAGES), and surgical header tolerance.

7. Forensic Capabilities Matrix & Verification

Module / Domain	Target Technologies	Key DFR-Forensics Capabilities	v2.8.0 Status
Disk Containers	RAW, DD, VMDK, VHD, VHDX, QCOW2, E01, AD1, AFF4, DMG	Strict read-only mounting, chunk-based multi-threaded I/O, virtual Copy-On-Write streams	Full Support
Partition Schemes	DOS MBR, Linked EBR, EFI GPT 2.10 Primary & Backup	Automated GPT restoration, IEEE 802.3 CRC32 recalculation, Protective MBR synthesis	Bit-Exact Repair
Windows Systems	FAT12/16/32, exFAT, NTFS (\$MFT, \$MFTMirr, \$LogFile), VSS	Orphan BPB derivation, \$MFTMirr failover, VSS snapshot timeline discovery, File Slack Inspector	Advanced + VSS + Slack
Linux Systems	EXT2, EXT3, EXT4 (Extents 0xF30A), CPIO, SquashFS v4	Sparse superblocks, double-indirect defragmentation, directory slack undelete, SquashFS reader	Advanced + SquashFS + CPIO
Apple Ecosystem	HFS+ / HFSX, APFS (Apple File System)	HFS B-Tree Catalog, \$AllocationFile bitmap, Object Map OMAP, separated APFS sub-volumes	New v2.8.0 (HFS+ & APFS)
Embedded & Flash	QNX4, QNX6, QNX F3S / ETFS, F2FS, EROFS, UBI / UBIFS	Raw flash erase units, in-place versioning, flash superblock sweep, 0xFF unprogrammed handling	New v2.8.0 (F3S / Flash)
Search & Carving	Multi-format carving, Raw Search, Binwalk Entropy	Multi-threaded streaming regex/Unicode search, CFTT de-braiding, strict frame validation	Complete + NIST CFTT 100%

Conclusion & Outlook

With version **v2.8.0**, **DFR-Forensics** establishes a unified forensic framework bridging conventional storage drives, modern client workstations, and mission-critical embedded systems. Through automated VSS snapshot discovery, surgical File Slack space inspection, bitmap-guided carving (\$AllocationFile / \$Bitmap), QNX F3S and SquashFS embedded parsing, and multi-threaded streaming raw search, the platform equips law enforcement agencies, incident response teams (CSIRT), and reverse-engineering labs with an uncompromisingly rigorous, transparent, and standards-compliant forensic investigative suite.